SUWAJ ARYAL

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OBJECTIVE

Civil and Hydropower Engineer with a strong foundation in advanced hydrological modeling, geospatial analysis, and computational hydrology. Seeking a PhD position to utilize my expertise in developing predictive models for complex catchments, with the goal of contributing novel research to the fields of hydro informatics and large-scale water resource management.

EDUCATION

MASTER OF SCIENCE IN HYDROPOWER ENGINEERING *(Ongoing)*
Pulchowk Campus, Tribhuvan University — 2024 – Present

BACHELOR OF CIVIL ENGINEERING
Pulchowk Campus, Tribhuvan University — 2018 – 2023

WORK EXPERIENCE

PRE-FEASIBILITY STUDY OF JUNBESI KHOLA CASCADE HYDROPOWER PROJECT (990kW) *(Freelance Engineer)*

- Design of intake, headrace pipe, weir, syphon and settling basin
- Design of anchor blocks and support pier
- Design of Turbine and powerhouse.

HYDRAULIC MODELLING OF BAGMATI RIVER *(Freelance Engineer)*

- 1D and 2D Hydraulic modelling of Bagmati river flow analysis for Urban area planning

INTERSHIP AT NCER

- Basics Hydraulic Modelling using HEC-RAS, HEC-HMS, EPA SWMM
- Automation scripts using VBA
- Drainage System Design

CIVIL DESIGN ENGINEER, MADI CONSULTANCY PVT. LTD.

- Irrigation and Water Supply Distribution Design

ACADEMIC PROJECTS

PRE-FEASIBILITY STUDY OF BADIGAD HYDROPOWER PROJECT *(Master's Minor Project)*

- Evaluated hydrological and topographical feasibility for run-of-river hydropower plant
- Contributed to layout planning and structural sizing of key components (intake, headrace tunnel, surge shaft)

PRELIMINARY ASSESSMENT OF RESERVOIR SEDIMENT MANAGEMENT STRATRGIES

- Reservoir Sediment Planning and management
- Using RESCON 2

KULEKHANI DAM HYDROLOGIC EXAMINATION

- Seepage analysis of Kulekhani Gravity Dam
- Hydropower station and Turbine location Visit

DETAIL DESIGN OF BHUTENI IRRIGATION PROJECT *(Bachelor's Final Year Project)*

- Performed detail design calculations for Bhuteni Irrigation Project.
- Hydraulics Structure such as Barrage, Under sluice, Canal, Cross Drainage structure (culvert), Canal and Bridge.

CONFERENCE

- Selection for conference presentation at CEISD-2026, 26th April 2026.
- "Suspended Sediment Concentration and Load Prediction using Data Driven Approach"

TRAINING

HYDRAULIC DESIGN SOFTWARE TRAINING (2-WEEKS)

- HEC-RAS, HEC-ResSim and HEC-HMS
- SWAT (Soil Water Assessment Tool)

TECHNICAL SKILLS

- Hydrological Tools: HEC- HMS, HEC- RAS, SWAT
- CAD Software: Civil 3D, AutoCAD
- GIS Software: QGIS, ArcGIS
- Data Base: Google Earth Engine
- Programming language: Python, JavaScript, HTML, C-programming Language, FORTAN, VBA
- CFD Software: Open FOAM
- MS Office (Word, Excel, PowerPoint)

STRENGTHS

- Strong academic foundation in civil and hydropower engineering principles
- Strong aptitude for numerical modelling and hydraulic analysis
- Research Enthusiastic
- Eager to learn and contribute in a multidisciplinary environment